

Product Quotient C^* -Algebras Fail the Ball-Covering Property

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June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when its pure state space, i.e. K , has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Let (E_n) be a sequence of nonzero normed spaces. Write

$$\ell_\infty(E_n) = \{x = (x_n) : \sup_n \|x_n\| < \infty\}$$

and

$$c_0(E_n) = \{x = (x_n) : \|x_n\| \rightarrow 0\}.$$

For $x \in \ell_\infty(E_n)$, denote its quotient class by $[x]$. The quotient norm is

$$\|[x]\| = \limsup_{n \rightarrow \infty} \|x_n\|.$$

Theorem 1. *For every sequence (E_n) of nonzero normed spaces,*

$$\ell_\infty(E_n)/c_0(E_n)$$

fails BCP.

Corollary 1. *Let (A_n) be a sequence of nonzero C^* -algebras. Then the C^* -algebra*

$$\prod_{n=1}^{\infty} A_n / \bigoplus_{n=1}^{\infty} A_n$$

fails BCP. In particular, for every nonzero C^ -algebra A , the sequence-algebra quotient $\ell_{\infty}(A)/c_0(A)$ fails BCP.*

Corollary 2 (quotient instability). *BCP and UBCP are not preserved by C^* -quotients. More precisely, if A is any nonzero separable C^* -algebra, then $\ell_{\infty}(A)$ has UBCP, but its quotient $\ell_{\infty}(A)/c_0(A)$ fails BCP.*

Idea of the Proof

This is the diagonal escape proof for ℓ_{∞}/c_0 , but with arbitrary nonzero fibers. Given countably many proposed centers in the quotient, choose for each center infinitely many coordinates where it nearly attains its quotient norm. On the selected coordinates, choose a unit vector in the fiber that is far enough from the corresponding center coordinate. The resulting bounded sequence has quotient norm one, while every proposed center sees infinitely many coordinates on which the distance is at least its own norm.

Proof

We use the standard equivalent form of BCP: a normed space Y has BCP iff there are countably many centers $y_k \in Y$ such that

$$S_Y \subseteq \bigcup_{k=1}^{\infty} B(y_k, \|y_k\|).$$

Lemma 1 (one-coordinate escape). *Let $E \neq \{0\}$ be a normed space and let $y \in E$. For every $\varepsilon > 0$, there is $u \in S_E$ such that*

$$\|u - y\| \geq \|y\| + 1 - \varepsilon.$$

In particular $\|u - y\| \geq \|y\|$ for a suitable choice of u .

Proof. If $y = 0$, take any $u \in S_E$. If $y \neq 0$, choose $f \in S_{E^*}$ with $f(y) = \|y\|$. Since $\|f\| = 1$, after changing phase if necessary one can pick $u \in S_E$ with $\operatorname{Re} f(u) < -1 + \varepsilon$. Then

$$\|u - y\| \geq \operatorname{Re} f(y - u) > \|y\| + 1 - \varepsilon.$$

Taking, say, $\varepsilon = 1/2$ gives the final assertion. □

Lemma 2. *There is a map $\pi : \mathbb{N} \rightarrow \mathbb{N}$ such that for every i and every N , some $k \geq N$ satisfies $\pi(k) = i$.*

Proof. List $\mathbb{N} \times \mathbb{N}$ as a sequence (i_k, j_k) , and put $\pi(k) = i_k$, arranging the list so each first coordinate occurs infinitely often. □

Proof of Theorem 1. Let $\xi_i = [a^i]$, $i \geq 1$, be any countable family in $\ell_{\infty}(E_n)/c_0(E_n)$, with representatives

$$a^i = (a_n^i)_{n \geq 1}.$$

We construct $b = (b_n) \in \ell_\infty(E_n)$ such that $\|b\| = 1$ and

$$\|b\| - \xi_i \geq \|\xi_i\|, \quad i \geq 1.$$

For each i , choose an increasing sequence $(n_{i,r})_{r \geq 1}$ such that

$$\|a_{n_{i,r}}^i\| \rightarrow \|\xi_i\|.$$

Let π be as in Lemma 2. We recursively choose a strictly increasing sequence of coordinates m_k . Having chosen m_{k-1} , choose $m_k = n_{\pi(k),r}$ for some large r so that

$$m_k > m_{k-1} \quad \text{and} \quad \|a_{m_k}^{\pi(k)}\| > \|\xi_{\pi(k)}\| - 1/k$$

whenever $\|\xi_{\pi(k)}\| > 0$. If $\|\xi_{\pi(k)}\| = 0$, only require $m_k > m_{k-1}$. In all cases m_k is chosen from the fixed subsequence $(n_{\pi(k),r})_r$.

At coordinate m_k , apply Lemma 1 in the nonzero space E_{m_k} to choose $b_{m_k} \in S_{E_{m_k}}$ such that

$$\|b_{m_k} - a_{m_k}^{\pi(k)}\| \geq \|a_{m_k}^{\pi(k)}\|.$$

For all other coordinates, set $b_n = 0$. Then $\|b\|_{\ell_\infty} = 1$, and because the selected coordinates are infinite, $\|b\| = 1$.

Fix i . Since $\pi(k) = i$ for infinitely many k , along that infinite subsequence the selected coordinates m_k are drawn from $(n_{i,r})_r$ with $r \rightarrow \infty$, and hence

$$\|a_{m_k}^i\| \rightarrow \|\xi_i\|.$$

Together with

$$\|b_{m_k} - a_{m_k}^i\| \geq \|a_{m_k}^i\|$$

this gives

$$\|b\| - \xi_i = \limsup_n \|b_n - a_n^i\| \geq \|\xi_i\|.$$

Thus $[b] \in S_{\ell_\infty(E_n)/c_0(E_n)}$ is in none of the open balls $B(\xi_i, \|\xi_i\|)$. Since the countable family (ξ_i) was arbitrary, the quotient fails BCP. $\square$

Corollary 1 follows by taking $E_n = A_n$ with its C^* -norm. The quotient $\prod A_n / \bigoplus A_n$ is exactly $\ell_\infty(A_n)/c_0(A_n)$ as a Banach space.

Proof of Corollary 2. It remains only to prove that $\ell_\infty(A)$ has UBCP when A is separable and nonzero. Choose a countable dense set $D \subseteq S_A$. For each $n \in \mathbb{N}$ and $d \in D$, define $c^{n,d} \in \ell_\infty(A)$ by

$$c_j^{n,d} = \begin{cases} 4d, & j = n, \\ 0, & j \neq n. \end{cases}$$

Then $\|c^{n,d}\| = 4$, so every ball $B(c^{n,d}, 7/2)$ avoids the origin. Let $x = (x_j) \in S_{\ell_\infty(A)}$. Pick n with $\|x_n\| > 3/4$, put $\alpha = \|x_n\|$, and set $u = x_n/\alpha$. Choose $d \in D$ with $\|d - u\| < 1/64$. At the n -th coordinate,

$$\|x_n - 4d\| \leq 4 - \alpha + 4\|u - d\| < 4 - \frac{3}{4} + \frac{1}{16} < \frac{7}{2}.$$

At all other coordinates, $\|x_j - c_j^{n,d}\| = \|x_j\| \leq 1 < 7/2$. Hence $\|x - c^{n,d}\| < 7/2$, and the countable family of equal-radius balls $B(c^{n,d}, 7/2)$ covers $S_{\ell_\infty(A)}$. Thus $\ell_\infty(A)$ has UBCP, while Corollary 1 says $\ell_\infty(A)/c_0(A)$ fails BCP. $\square$

Scope and Relation to the C*-Algebra Question

This gives a broad negative family for the source question. It does not characterize arbitrary C^* -algebras, but it shows that sequence-algebra coronas are robustly outside BCP. In particular, even when the fibers are separable or finite dimensional, the product quotient has too much tail room for any countable ball cover to capture its unit sphere. The quotient instability corollary is also a warning: a C^* -algebraic characterization of BCP cannot be based on quotient-stable invariants alone.

Novelty and Search Bounds

The run indexes were searched for arXiv:2311.14257, product quotients, ℓ_∞/c_0 , Calkin/corona algebras, and BCP. Web and local source checks found the classical scalar obstruction and the later Calkin-algebra paper [2], but no exact product-of-arbitrary-fibers statement in the run memory. The theorem is a direct generalization of the scalar ℓ_∞/c_0 diagonal construction.

Human Review Focus

The proof is elementary. Review the one-coordinate escape lemma and the diagonal construction of the coordinates m_k , especially the fact that each center i is attacked along infinitely many coordinates where its representative nearly realizes the quotient norm.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Qiyao Bao, Rui Liu, and Jie Shen, *The ball-covering property of non-commutative spaces of operators on Banach spaces*, arXiv:2412.07137, 2024.